

Duolingo

NASDAQ: DUOL
Current Price: $ 95.18
Target Price: $ 157.1
Total 24-months return: 65%
Recommendation: LONG

Jayden Fung and Armaan Niaz

Duolingo's priced on negative sentiment, not strong fundamentals

A profitable learning leader with sticky subscribers and strong growth priced at mature-consumer multiples

Thesis Overview

High-quality business with strong execution history

- Freemium subscription-based business model creates predictable cashflow
- A strong time moat solidifies Duolingo's position as the digital learning leader

Attractive valuation due to poor sentiment

- AI concerns & pivots to DAU growth instead of monetization have resulted in market's being spooked
- Duolingo is trading at near all time lows, undervalued against peers

Key Stats

Currency	USD
Trading	
Current Price	$95.18
%1Y Chg	-66.55%
52-wk range	$91.99 - $544.93
Shares Outstanding, Diluted	49.78
EPS Diluted	$11.88
TTM P/E	11.75
Forward P/E	12.35
Capitalization	
Market Cap	$4,738.06
+ Debt	$93.78
- Cash	$1,036.39
Enterprise Value	**$3,795.45**

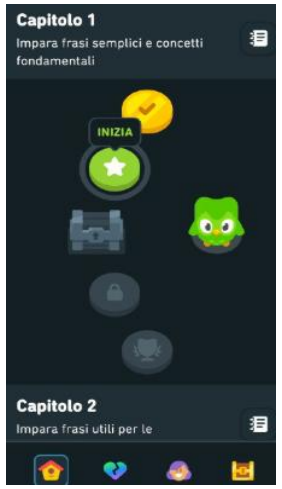
(all values in millions, except per share values)

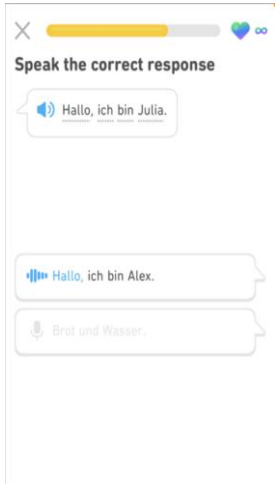
Source: Bloomberg, Yahoo Finance, Visible Alpha

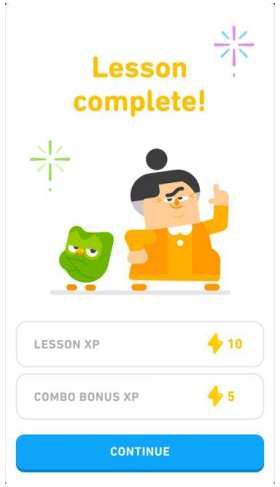
Duolingo at a Glance

A gamified education app turning learning into a daily habit

Duolingo is a gamified digital education app primarily focused on language learning across **40+ languages**, and has recently expanded into **Chess, Math**, and **Music**. It teaches **short, game-like lessons** featuring **XP**, **streaks**, **leaderboards**, through its iconic owl mascot **Duo** and characters which hook users and keep them returning daily







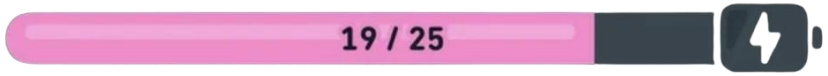




Duo, Duolingo's green owl mascot, is central to the brand's identity. Beyond serving as the face of the product, Duo drives significant awareness through being the face of unhinged, and viral marketing content that drives Duolingo's **90% organic growth** that has made Duolingo one of the most recognisable consumer apps globally

Freemium Funnel Penetration

Free Version





Users only get 25 **Energy Points** (Questions per day) and are forced to watch ads to regain energy. Users are also hit with **unskippable ads** after lessons, pertaining to the benefits of Super which include **Unlimited Energy**, **No Ads**, **Full Offline Access** and **Streak Protection** which induces subscription upgrades

Duolingo Super

Roleplay



$83.99/ year

Users on Super are usually highly invested into their learning experience, which incentivize them to move past text-based learning and onto Duolingo's conversational tools: **Roleplay** and **Video Call**

Video Call



Duolingo Max

$167.99/ year

Max users are retained through constant in-app visibility of **locked Max features**, which means that any user who downgrades is repeatedly reminded of the features they've lost, and cancelling Max erases personal conversational records, which raises the perceived cost of leaving and increases re-upgrade pressure

High Quality Business

Strong historical growth accompanied by a large, growing industry

Strong FY25 Fundamentals

- 39% revenue growth where 84% is from recurring subscriptions
- Strong userbase growth, with 2.7M new paid subscribers
- CAC at ~$9.3 per net daily active user added
- Strong ARPU growth demonstrates growth by Duolingo Max, their premium AI-powered
- New expansion into chess reaching 7M DAU

Many Market Opportunities

18% growth in Online Language Learning

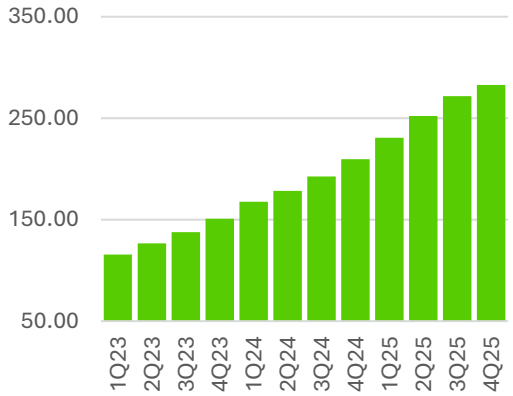
- The online language learning market expected to grow at a ~18% CAGR to $91.6B (from $33B in 2024)
- Sustained growth in mature markets & growing customer base in newer markets like China,

Large TAM where doubling Daily Active Users is realistic

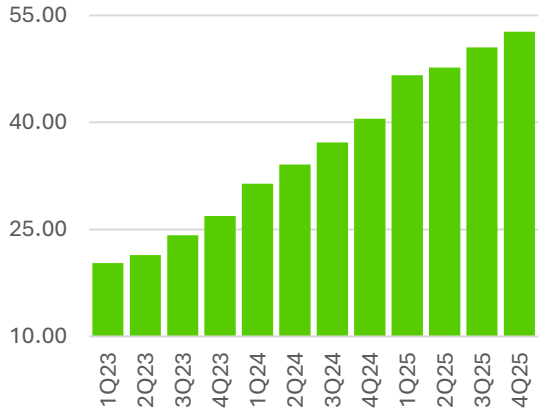
- Management guidance of 100M DAU by 2028
- 2% of U.S. internet users use Duolingo every day
- HolonIQ estimates market total users at 2B

Key Financial Metrics (FY25)

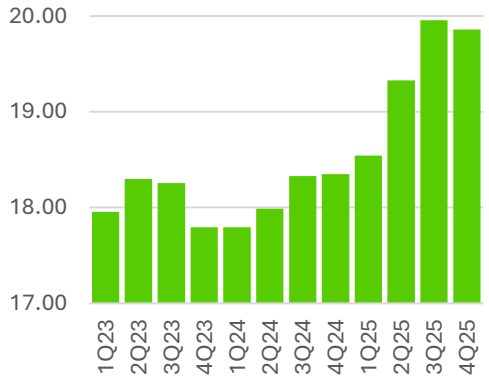
Revenue ($M)



Daily Active Users (M)



Average revenue per paid subscriber ($)



Daily Active Users
52.7M
+30% yoy

Monthly Active Users
131.7M
+22% yoy

Paid Subscribers
12.2M
+28% yoy

TAM Source: https://uk.finance.yahoo.com/news/online-language-learning-generate-91-162300022.html

All figures for FY25 / 4Q25

Strong Moat through habits and data instead of features

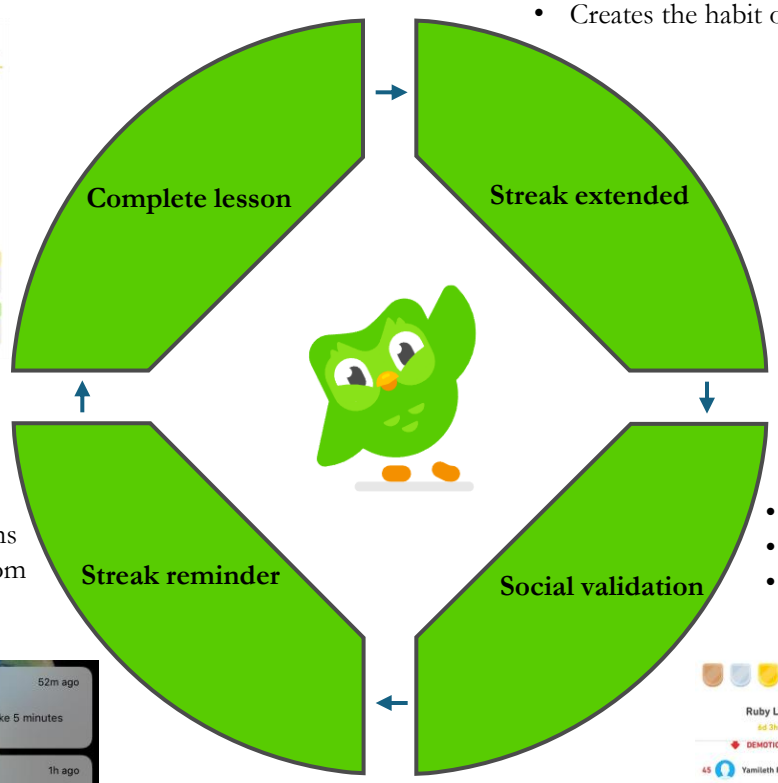
Building features ≠ Building Habits – The moat is behavioural

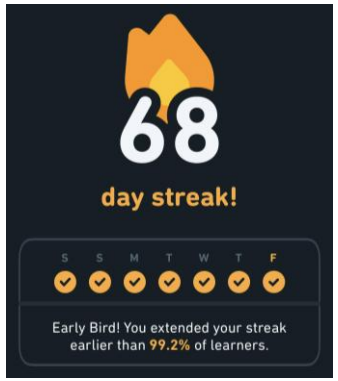
The Gamification Cycle & Habit formation

- 15 minute bite-sized commitment
- Instant gratification

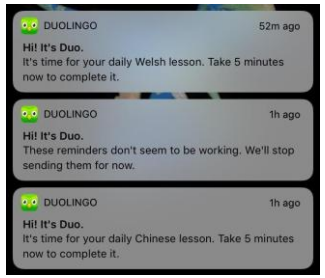
- Larger psychological cost to quit from longer streaks
- Creates the habit of using Duolingo every day



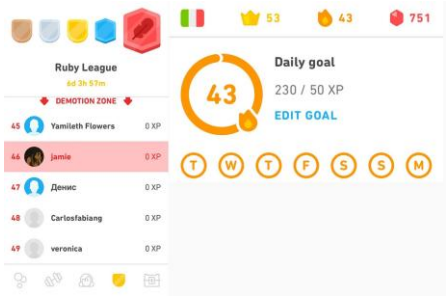




- Push notifications
- Friendly guilt from Duo the owl



- Leaderboard ranking updated
- Friends can see progress
- Weekly leagues for competition



What Duolingo has that others don't

Network effects can't be replaced by AI

- Friend challenges and family leagues create social accountability to continue learning on Duolingo
- Duolingo social media & user-generated content create community interaction, increasing engagement
- Gamified social features (e.g. friend quests) have users work together towards a goal, driving virality

An abundance of data leads to consistent experimentation

- 50M+ DAUs provide massive sample sizes that no other competitor has
- Can test tiny UI changes with statistical significance, allowing for percentage points worth of retention gains
- These gains slowly compound over time, creating a product that beats competition

Character IP & Brand Affinity lowers CAC, increases retention

- Duolingo's Green Owl acts as a recognisable and emotionally engaging character, with 10%+ of users saying it directly motivates their app usage
- Strong brand affinity enables monetisation of gamified features (streak protection, leagues, Energy, Max) without hurting engagement

Data enables lower churn, more habit formation

Usage Drivers (Oct 25')

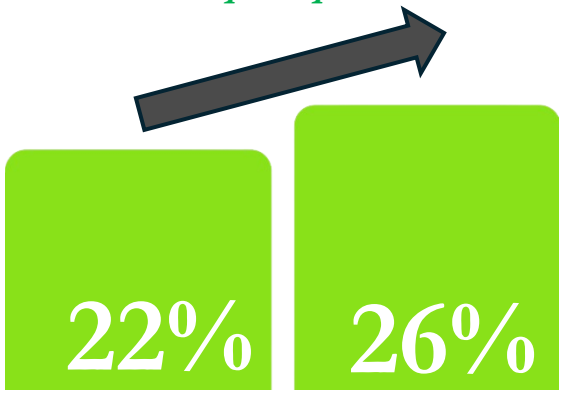


70% of users are driven by Duolingo's **Notifications** and **Daily Streaks**

Engagement into Routine

% of Users with 90+ day streak

400 bps expansion

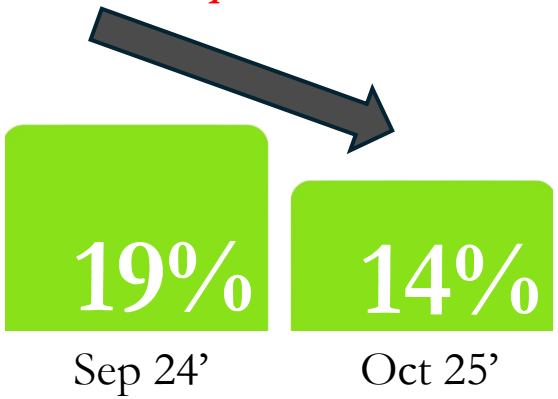


Growth in 90+ day streaks reflects users progressing beyond casual use into persistent routines, highlighting the effectiveness of streaks and Duolingo's gamification in driving long-term retention.

Early-stage churn is declining

% of Users with <7-day streak

500 bps contraction

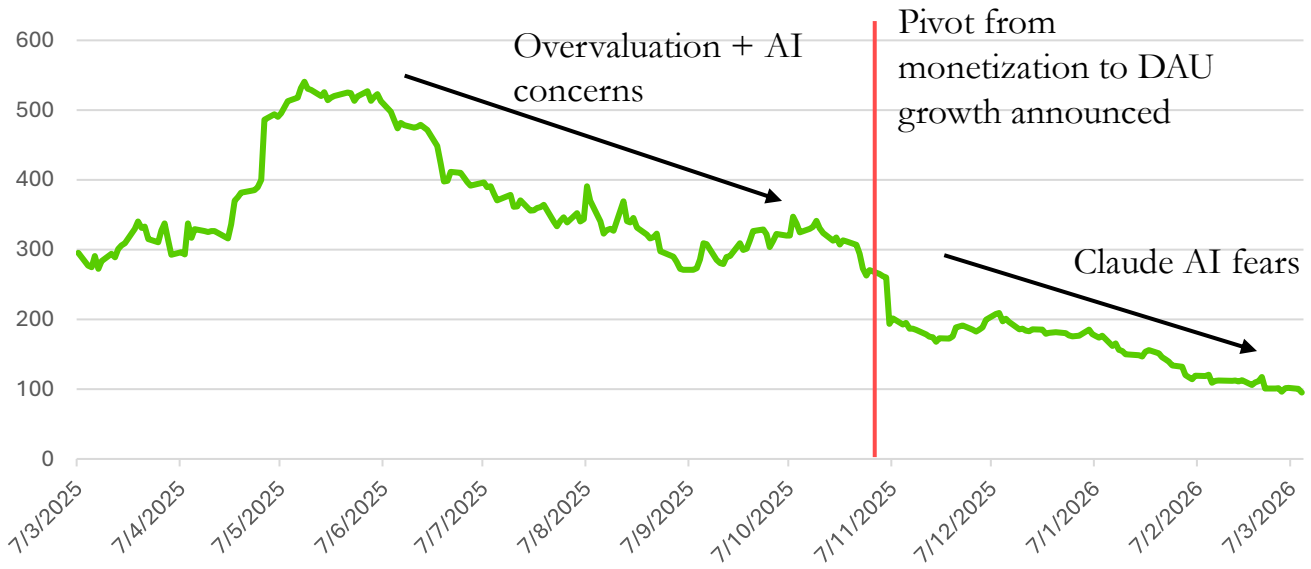


The reduction in sub-7-day streak users suggests improved onboarding effectiveness and lower initial drop-off, which increases the probability that new and returning users progress into longer engagement cohorts.

Strong fundamentals, persistent 'Buy' ratings, yet the stock declines?

Let's see what's happened over the past 2 years

Misguided sentiment has resulted in valuation compression



Comps analysis

	Forward P/E	Forward Revenue Growth
Pearson	15.75	4.27%
新东方 XDF.CN NEW ORIENTAL EDUCATION & TECHNOLOGY GROUP	12.64	11.73%
STRATEGIC EDUCATION INC	11.04	2.54%

Why the AI argument is unjustified

"AI translation has been essentially perfect among the large languages for like more than 10 years. I mean between Spanish and English, it's essentially been perfect. Our users use Duolingo for 2 main reasons: one, as a hobby, they actually want to learn. And those people, whether there's AI translation, it doesn't matter, it's a hobby for them. And then the other group of people is learning English. They want to learn English, and they actually want to learn English." – Luis von Ahn, CEO (Q4 2025 Earnings Call)

Our valuation – multiple compression from sentiment isn't fundamental deterioration

Key Variant Items 2025 → 2028	BEAR	BASE	BULL
DAU	75m	80m	85m
DAU to Paid Subscriber Conversion Rate	21.3%	21.3%	21.3%
ARPPS	$71.6	$71.6	$71.6
Net Profit Margin	27.2%	27.4%	27.7%
Exit P/E	11x	13x	16x
Target Price	$110.8	$157.1	$171.9
Total 2 Year Return	16.37%	65.1%	80.6%

Potential catalysts and risks

Short-Term Catalysts

Q1 / Q2 bookings guidance beats

Guidance above expectations would show that Duolingo's focus on engagement is not hurting revenue growth, reducing fears of a 20% growth slowdown in early 2026

Increase in engagement from Duolingo Chess / Maths

Helps show Duolingo's long-term vision to expand into all education sectors

Medium-Term Catalysts

Greater international expansion

Rising adoption outside the U.S., particularly in English-learning markets, expands the pool of future paying users, with Max approval in China providing incremental growth

Sentiment Recovery

As Duolingo continues to grow, investors will see how AI can't easily replace Duolingo, and how DAU expansion helps the business in the long run

Increased DET Revenue

Growth in DET volumes adds a high-margin, non-subscription revenue stream that requires little incremental marketing spend.

Risk #1: Guidance Conservatism

Management's more cautious guidance approach risks anchoring investor expectations to sub-20% bookings growth even if underlying engagement and monetisation trends remain healthy

Mitigation: Despite recent conservative framing, Duolingo has a history of beating initial guidance, as management has consistently under-promised on bookings and profitability while benefiting from resilient DAU growth and gradual monetisation upside

Risk #2: Macro Headwinds

A softer macro environment may pressure discretionary spending, increasing resistance to subscription upgrades, which limits Duolingo's short-term ARPU expansion

Mitigation: Duolingo's freemium model and low price point relative to traditional courses make it a resilient affordable luxury and its strong habit forming mechanics further insulates retention.

APPENDIX

DCF VALUATION (FYE Dec 31)

Base Case	Actuals		Projections			Fade to Terminal						
	2024A	2025A	2026E	2027E	2028E	2029E	2030E	2030E	2031E	2032E	2033E	2034E
Revenue	748	1,038	1,150	1,408	1,650	1,914	2,182	2,444	2,688	2,903	3,106	3,293
% yoy		38.77%	10.79%	22.43%	17.19%	16.0%	14.0%	12.0%	10.0%	8.0%	7.0%	6.0%
EBIT	190	296	282	362	424	479	545	611	672	726	777	823
% margin	25.40%	28.52%	24.53%	25.73%	25.68%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%
Less: Taxes	57	86	92	113	132	152	175	201	231	265	305	351
% rate	25.00%	25.00%	25.00%	25.00%	25.00%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%
NOPAT	133	210	190	250	292	327	371	410	441	460	471	472
(+) SBC	120	149	168	203	224	268	305	342	376	406	435	461
% of sales	16.04%	14.35%	14.57%	14.40%	13.60%	14.0%	14.0%	14.0%	14.0%	14.0%	14.0%	14.0%
(+) D&A	11	14	13	15	17	21	24	27	30	32	34	36
% of sales	1.47%	1.35%	1.14%	1.08%	1.01%	1.1%	1.1%	1.1%	1.1%	1.1%	1.1%	1.1%
(-) Capex	12	18	18	21	22	27	31	34	38	41	43	46
% of sales	1.60%	1.73%	1.56%	1.49%	1.32%	1.4%	1.4%	1.4%	1.4%	1.4%	1.4%	1.4%
(-) Changes in NWC	71	53	39	52	66	77	87	98	108	116	124	132
% of sales	9.49%	5.11%	3.37%	3.69%	4.01%	4.0%	4.0%	4.0%	4.0%	4.0%	4.0%	4.0%
Unlevered Free Cash Flow (UFCF)	**181**	**302**	**314**	**395**	**445**	**512**	**583**	**647**	**702**	**742**	**773**	**791**
% yoy		66.85%	4.00%	25.66%	12.71%	15.18%	13.70%	11.10%	8.45%	5.70%	4.14%	2.44%
% margin		**29.09%**	**27.31%**	**28.03%**	**26.96%**	**26.77%**	**26.70%**	**26.49%**	**26.11%**	**25.56%**	**24.87%**	**24.04%**
Discount period			1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00
PV of UFCF			294	347	366	395	422	439	446	442	432	414
Cumulative PV of UFCF			3,998									

Equity Valuation (As on 10/3/26)	
Enterprise Value	9,927
Total Debt	94
(-) Cash	-1,141
Net Debt	(1,047)
Equity Value	10,974
FDSO	50
Implied share price	**219.49**
Current share price	95.18
Implied upside/(downside)	130.6%

WACC (FY25, as on 10/3/26)	
Equity	
% of equity	98.50%
Cost of equity	6.73%
Risk free rate	4.50%
Beta	0.89
Market risk premium	7.00%
Debt	
% of debt	1.50%
Cost of debt	5.50%
Tax rate	25.00%
WACC	6.69%

Terminal Value	
Terminal Value (Gordon's Growth Method)	14,059
% terminal growth rate	1.0%
PV of TV	7,360
% weight	50.0%
Terminal Value (Exit Multiple Method)	8,594
Terminal year EBITDA	859
Exit multiple	10.0x
PV of TV	4,499
% weight	50.0%
Weighted PV of TV	5,930

DCF VALUATION (FYE Dec 31)

	Actuals		Projections			Fade to Terminal						
Bear Case	2024A	2025A	2026E	2027E	2028E	2029E	2030E	2030E	2031E	2032E	2033E	2034E
Revenue	748	1,038	1,150	1,408	1,650	1,733	1,819	1,910	2,006	2,106	2,211	2,322
% yoy		38.77%	10.79%	22.43%	17.19%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%
EBIT	190	296	282	362	424	433	455	478	501	526	553	580
% margin	25.40%	28.52%	24.53%	25.73%	25.68%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%
Less: Taxes	57	86	92	113	132	152	175	201	231	265	305	351
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NOPAT	133	210	190	250	292	281	280	277	271	261	247	229
(+) SBC	120	149	168	203	224	243	255	267	281	295	310	325
% of sales	16.04%	14.35%	14.57%	14.40%	13.60%	14.0%	14.0%	14.0%	14.0%	14.0%	14.0%	14.0%
(+) D&A	11	14	13	15	17	19	20	21	22	23	24	26
% of sales	1.47%	1.35%	1.14%	1.08%	1.01%	1.1%	1.1%	1.1%	1.1%	1.1%	1.1%	1.1%
(-) Capex	12	18	18	21	22	24	25	27	28	29	31	33
% of sales	1.60%	1.73%	1.56%	1.49%	1.32%	1.4%	1.4%	1.4%	1.4%	1.4%	1.4%	1.4%
(-) Changes in NWC	71	53	39	52	66	69	73	76	80	84	88	93
% of sales	9.49%	5.11%	3.37%	3.69%	4.01%	4.0%	4.0%	4.0%	4.0%	4.0%	4.0%	4.0%
Unlevered Free Cash Flow (UFCF)	**181**	**302**	**314**	**395**	**445**	**449**	**457**	**462**	**465**	**465**	**462**	**455**
% yoy		66.85%	4.00%	25.66%	12.71%	1.02%	1.62%	1.18%	0.66%	0.04%	(0.71%)	(1.61%)
% margin		**29.09%**	**27.31%**	**28.03%**	**26.96%**	**25.94%**	**25.10%**	**24.19%**	**23.19%**	**22.09%**	**20.89%**	**19.58%**
Discount period			1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00
PV of UFCF			294	347	366	347	330	313	296	277	258	238
Cumulative PV of UFCF			3,067									

Equity Valuation (As on 10/3/26)	
Enterprise Value	6,269
Total Debt	94
(-) Cash	-1,141
Net Debt	(1,047)
Equity Value	7,316
FDSO	50
Implied share price	**146.31**
Current share price	95.18
Implied upside/(downside)	53.7%

WACC (FY25, as on 10/3/26)	
Equity	
% of equity	98.50%
Cost of equity	6.73%
Risk free rate	4.50%
Beta	0.89
Market risk premium	7.00%
Debt	
% of debt	1.50%
Cost of debt	5.50%
Tax rate	25.00%
WACC	6.69%

Terminal Value	
Terminal Value (Gordon's Growth Method)	7,384
% terminal growth rate	0.5%
PV of TV	3,866
% weight	50.0%
Terminal Value (Exit Multiple Method)	4,848
Terminal year EBITDA	606
Exit multiple	8.0x
PV of TV	2,538
% weight	50.0%
Weighted PV of TV	3,202

Credible, experienced management reduces execution risk as Duolingo scales



Luis Von Ahn
CEO (Aug 2011 - Current)

Founder with a track record of building products that change user behaviour at global scale (reCAPTCHA), directly supporting Duolingo's long-term habit-based growth model

His computer science and AI background means he knows how to implement AI features into the app (AI personalisation, AI content creation) which helps retention and learning improve

He is Closely involved in how new subjects are built, making sure English, Math, Music, and Chess all run on the same engagement and experimentation engine rather than drifting into separate products



Gillian Munson
Incoming CFO (Feb 2026)

Deep institutional knowledge from 6+ years on Duolingo's Board gives Munson continuity as CFO, allowing faster capital allocation decisions and reducing execution risk as the company scales AI, pricing, and new verticals

Board-level experience at Phreesia strengthens her governance and long-term decision-making discipline, directly relevant as Duolingo balances growth investment with increasing profitability and public-market scrutiny

CFO experience at Vimeo, is directly applicable to Duolingo's challenge of converting engagement into revenue without hurting retention, particularly as Duolingo expands premium tiers (Super, Max, Family)

Extra JPMorgan survey information

Figure 2: Which of the following best describes your use case? (Fall 2025)

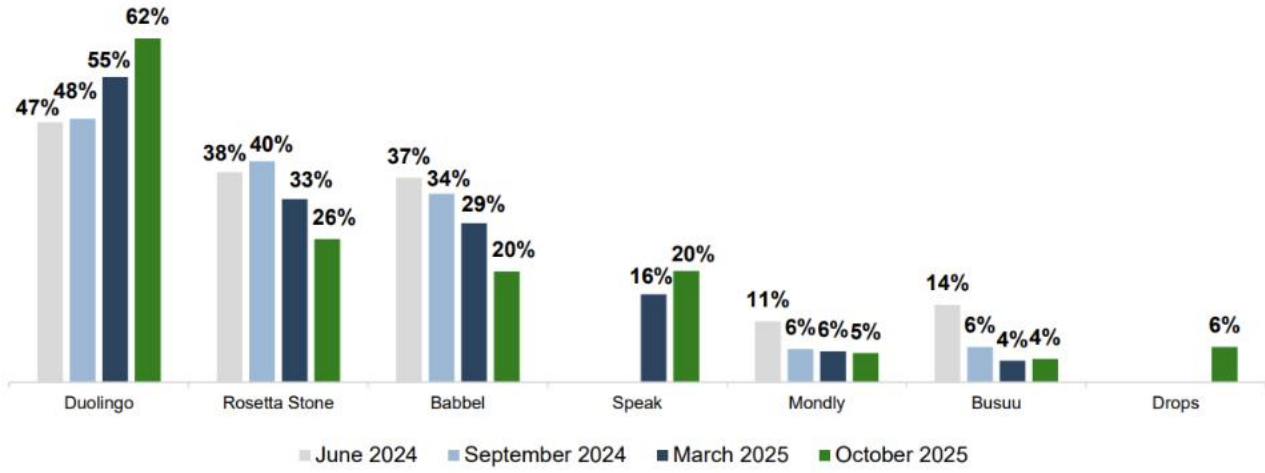
Answers	TOTAL
Casual Learner / Brain Training	**33%**
Travel / Social	**23%**
Family / Culture	**20%**
Work	18%
College Learning Supplement	4%
Other (please specify)	1%
TOTAL	100%

Source: JPM – Internet/Online Education Fall 2025 Survey (1,038 total responses). Figures may not add up to 100% due to rounding.

Figure 16: What drove you to start using Duolingo?

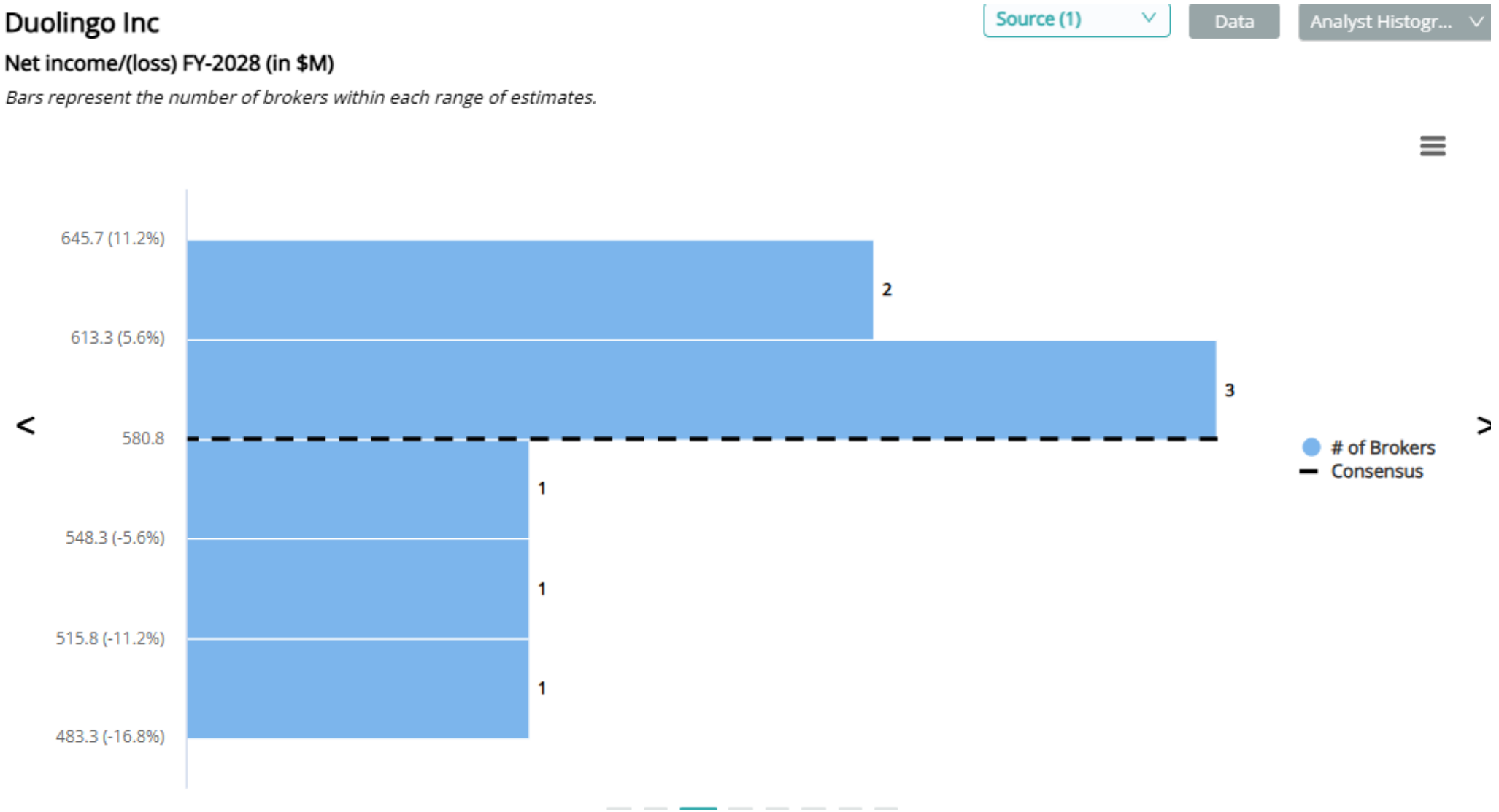
Answers	TOTAL
Friends/Family	**46%**
TikTok	**19%**
YouTube	14%
Instagram	8%
TV/Streaming	5%
Other (please specify)	5%
News/Article/Blog	3%
TOTAL	100%

Figure 9: Which Language Learning service(s) do you currently use? (Time Series)



Source: JPM – Internet/Online Education Surveys. Figures may not add up to 100% due to rounding.

VisibleAlpha Extra Consensus Data



VisibleAlpha Extra Consensus Data

Line Item	Units	FY-2025 E Consensus	FY-2026 E Consensus	FY-2027 E Consensus	FY-2028 E Consensus
Operating metrics					
Bookings					
Bookings - Subscription	$M	995.49	1,217.27	1,470.71	1,743.89
Bookings - Other segment	$M	160.80	176.99	184.28	211.23
Bookings	$M	1,156.06	1,394.25	1,654.99	1,955.11
Subscription Bookings / MAU ($)	$Abs	7.46	7.90	8.34	8.72
Revenue					
Revenue - Subscription	$M	867.55	1,082.42	1,311.85	1,565.16
Revenue - Other segment	$M	162.09	177.22	193.30	215.64
Total revenue	$M	1,030.98	1,259.61	1,505.11	1,780.72
User Metrics					
Monthly active users (MAU)	M#	133.47	154.47	177.39	200.50
Net adds - MAU	M#	24.60	21.00	22.75	23.45
Daily active users (DAU)	M#	49.34	61.40	74.33	85.39
Net adds - DAU	M#	13.54	12.07	12.90	12.57
Paid subscribers	M#	11.95	14.45	17.43	19.94
Net adds - Subscribers	M#	3.18	2.49	3.05	2.94
DAU to MAU	%	37.03	39.80	42.05	42.81

RDCF Analysis

Discounted Cash Flow Analysis	Magnitude	FY0 FY-2024	FY1 FY-2025	FY2 FY-2026	FY3 FY-2027	FY4 FY-2028	FY5 FY-2029	FY6 FY-2030	FY7 FY-2031	FY8 FY-2032	FY9 FY-2033	FY10 FY-2034
Net Revenue	M	748	1,031	1,062	1,094	1,127	1,160	1,195	1,231	1,268	1,306	1,345
Net Revenue % Growth			*37.8%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*
Gross Profit / (loss)	M	544	743	765	788	811	835	861	886	913	940	969
% of revenue		*72.8%*	*72.0%*	*72.0%*	*72.0%*	*72.0%*	*72.0%*	*72.0%*	*72.0%*	*72.0%*	*72.0%*	*72.0%*
Operating Income		63	128	159	164	169	174	179	185	190	196	202
% of revenue		*8.4%*	*12.4%*	*15.0%*	*15.0%*	*15.0%*	*15.0%*	*15.0%*	*15.0%*	*15.0%*	*15.0%*	*15.0%*
EBITDA	M	192	300	356	366	377	389	400	412	425	438	451
EBITDA Growth			*56.1%*	*18.7%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*
EBITDA Margin			*29.1%*	*33.5%*	*33.5%*	*33.5%*	*33.5%*	*33.5%*	*33.5%*	*33.5%*	*33.5%*	*33.5%*
Taxes		4	(101)	12	17	20	35	36	37	38	39	40
Tax Rate	Abs	6.0%	-78.7%	7.8%	10.6%	11.9%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%
Net Income	M	89	414	212	278	347	232	239	246	254	261	269
Net Income (%) Growth			*367.3%*	*-48.7%*	*30.7%*	*25.1%*	*-33.2%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*
Net Income Margin			*40.1%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*	*20.0%*
SBC	M	110	141	127	131	135	139	143	148	152	157	161
% of revenue		15%	14%	12%	12%	12%	12%	12%	12%	12%	12%	12%
Capex		21	15	18	18	19	23	24	25	25	26	27
Capex as % Sales	Abs	*2.8%*	*1.5%*	*1.7%*	*1.6%*	*1.7%*	*2.0%*	*2.0%*	*2.0%*	*2.0%*	*2.0%*	*2.0%*
Change in Working Capital	M	76	17	77	92	113	70	72	74	76	78	81
Change in Working Capital as % of Net Revenue		*10.1%*	*1.6%*	*7.2%*	*8.4%*	*10.0%*	*6.0%*	*6.0%*	*6.0%*	*6.0%*	*6.0%*	*6.0%*
	M	11	13	12	15	14						
Free Cash Flow	**M**	**202**	**509**	**376**	**370**	**361**	**400**	**412**	**425**	**437**	**451**	**464**
FCFF % Growth			***153%***	***-26%***	***-1%***	***-3%***	***11%***	***3%***	***3%***	***3%***	***3%***	***3%***
Diluted Shares Outstanding	M	47	49	50	52	55	59	57	61	63	65	67
FCFF Per Share		4.28	10.35	7.45	7.10	6.50	6.79	7.24	6.98	6.97	6.95	6.95
Discount Factor		*100%*	*93%*	*86%*	*79%*	*74%*	*68%*	*63%*	*58%*	*54%*	*50%*	*46%*
Present Value of FCFF		201.61	471.44	321.95	293.66	265.02	272.46	259.84	247.81	236.34	225.40	214.96

Calculated WACC	8.00%	*Increments*	0.50%
Perpetuity	464		
Discount Factor	46%		
Net Present Value	3,010		
FCFF Terminal Growth Rate	2.00%		
Terminal Value	3,583		
EV	6,593		
Net Debt	(823)		
Minority Interest	-	N	
Market Value	5,770		
Weighted average shares outstanding, Diluted	50	M	
Equity Value Per Share	115.78		
Stock Price (Prior Close)	111.74		
Upside / Downside	**3.62%**		

Price		WACC							
		6.50%	7.00%	7.50%	8.00%	8.50%	9.00%	9.50%	10.00%
Terminal Growth Rate	0.00%	114.31	108.17	102.71	97.81	93.38	89.35	85.65	82.25
	0.50%	119.84	112.91	106.82	101.41	96.55	92.17	88.18	84.52
	1.00%	126.37	118.44	111.56	105.51	100.15	95.34	91.00	87.04
	1.50%	134.22	124.97	117.09	110.25	104.26	98.93	94.17	89.86
	2.00%	143.80	132.82	123.62	115.78	109.00	103.04	97.76	93.03
	2.50%	155.78	142.40	131.47	122.32	114.53	107.78	101.87	96.63
	3.00%	171.19	154.38	141.05	130.16	121.06	113.31	106.61	100.74
	3.50%	191.73	169.79	153.03	139.75	128.91	119.85	112.14	105.48
	4.00%	220.49	190.33	168.44	151.73	138.49	127.69	118.68	111.01
	4.50%	263.62	219.09	188.98	167.14	150.47	137.28	126.52	117.54

The Green Machine: Six Steps

1. Staff It with Great People

Great experiments require exceptional teams who can quickly brainstorm ideas, execute, and pivot on a dime. This is why we set such a high bar in hiring and invest so much in our people. We like to keep teams lean and scrappy to start, only adding more people once we see success.

2. Define Success

We always want to set clear, measurable goals—whether it's DAUs, bookings, or social impressions. But metrics aren't always available for early-stage projects. In that case, we define the best available qualitative goals, iterate toward clarity, and keep an eye out for any useful signals.

3. Set Guardrails and Think Long-Term

We use guardrails to ensure our work benefits both Duolingo and our learners over the long term. This aligns with our core principle, Take the Long View. Thoughtful guardrails help us focus on sustainable, meaningful growth, ensuring that each experiment serves a greater purpose.

4. Build the Thing and Set Up Feedback Loops

The fastest way to create something excellent isn't through abstract discussions. Just start building, and create the right continuous feedback loops. This feedback comes from both quantitative (A/B test results, social impressions, efficacy research) and qualitative sources (extensive dogfooding). Early-stage projects typically lean more heavily on qualitative feedback before shifting to more data-driven metrics.

5. Execute with Urgency and Excellence

Because of the compounding nature of smaller improvements, there is a huge cost to moving slowly. But moving too fast can also lead to sloppy outcomes. In step five, we work to continuously (and urgently) ship work that meets our bar.

6. Double Down on What Works, Stop What Doesn't

When something is working, we allocate more resources and keep working on it (sometimes for years). But we need to be comfortable sunsetting projects or even entire products that don't demonstrate success.

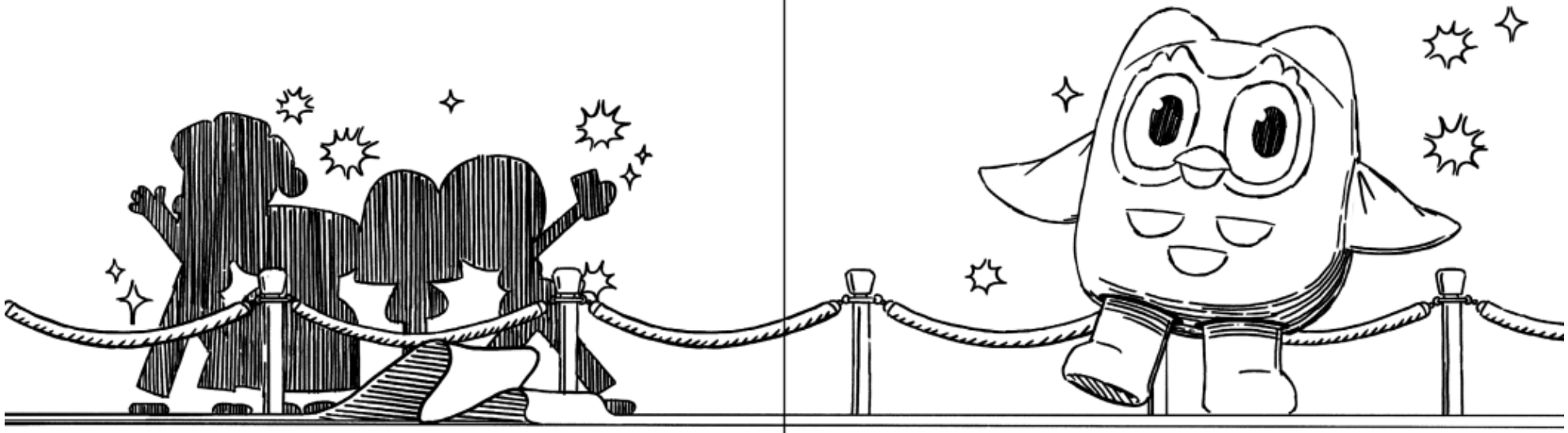
⚠ ***Part of the Plan***

Crucially, the Green Machine is not a replacement for planning. We still have quarterly OKRs, annual goals, and a long-range financial plan. The difference is that we don't rely on multi-year roadmaps for the app.

Characters Count

Our characters make learning fun. Built from simple geometric shapes, each has a distinct personality and story. Together, they reflect the diversity of our learners and bring a sense of humor and play to the app.

But these characters—inspired more by brands like Nintendo than traditional learning companies—also serve an important strategic purpose. We see this IP as a crucial moat for our business, especially in the era of AI learning tools. That same emotional connection that makes learning enjoyable also makes our product stickier over time. Even if someone cloned our entire app, learners would still come back to Duolingo for the characters.



Investing in Marketing

At Duolingo, we've grown by creating a product that people genuinely love—and love to talk about. But we've also figured out how to use other levers, like performance marketing, to complement that organic growth. Our paid ads exist to amplify our word-of-mouth momentum, not replace it, so we are conservative with our spend. This allows us to invest in standout campaigns and reactive social moments that set us apart.

Our marketing team excels at creating high-impact content at a low cost. Thanks to their work, Duo has walked the red carpet at the *Barbie* premiere, hosted a (fake) dating reality show on the streaming service Peacock, and become an international social media star. These moments reflect the playful, irreverent energy of our brand and show how a thoughtful balance of organic growth and marketing can create meaningful impact.

The Path

Originally, learners navigated Duolingo through the Tree, which consisted of skills (e.g., going to a restaurant). Each skill began with basic word meanings and progressed to harder content, but learners only needed to complete the first level to move on. While this flexibility let them advance quickly, it often led to frustration when harder concepts came along and exposed gaps in their foundational knowledge.

In 2022, we took a leap: replacing the Tree with the Path. This new linear structure requires learners to complete every lesson in sequence, ensuring a consistent and strong foundation for everyone. The Path was a large undertaking across both engineering and design. We anticipated that learners might resist such a dramatic change to their routine. And even though the data didn't immediately show improvements in core metrics like DAUs or monetization, the switch was the right decision. It prioritized real mastery over speedrunning easy lessons to maintain streaks. It also gave us clearer insights into learner progress and greater control over the learning experience.

Sometimes taking the long view means making bold moves that might seem counterproductive in the moment. The Path was just that. It fundamentally changed how people learn with Duolingo and has been essential to our success.



Building a Forever Product

USING DUOLINGO IS LIKE EATING BROCCOLI BUT TASTING ICE CREAM. YOU ENJOY IT — AND YOU GET SOMETHING VALUABLE OUT OF IT.

- BOZENA PAJAK

The Duolingo app is designed to be sticky in the short-term and transformative in the long-term. For years, we've focused on building something so useful and delightful that it becomes an essential part of our learners' lives.

Learning—particularly language learning—requires regular practice over extended periods of time. That's why we prioritize user retention, and have spent years perfecting the Streak feature. The more we can do to keep our learners committed for the long haul, the more value they'll get from Duolingo.

We also invest in teaching better. Even though this doesn't generate immediate revenue, we know that if a product fails to deliver on its promise, people will eventually stop using it.



Notifications

Every day, Duolingo sends millions of messages to learners. These notifications present a fundamental tension between long-term thinking and short-term goals.

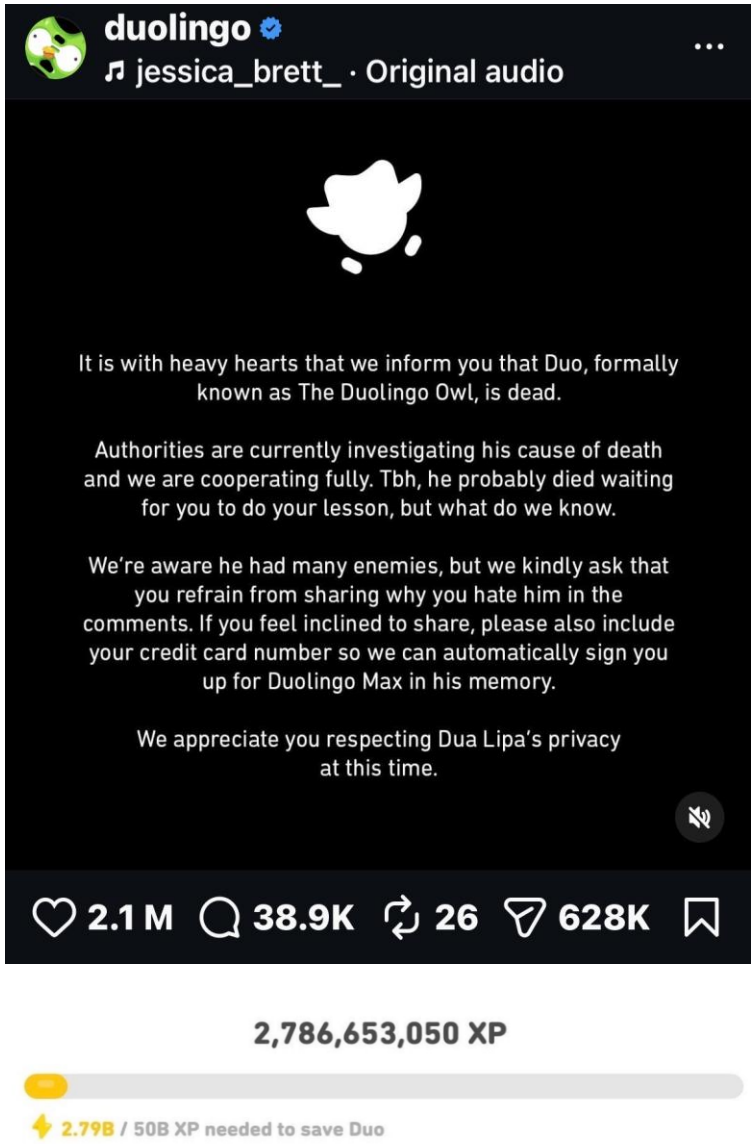
More notifications leads to more Daily Active Users (DAUs)* in the short-term. But people who feel bombarded don't stick around for long—they eventually turn off notifications or abandon the app entirely. That's why we put firm limits on notifications, regardless of what the short-term metrics suggest. User trust matters more than immediate gains.

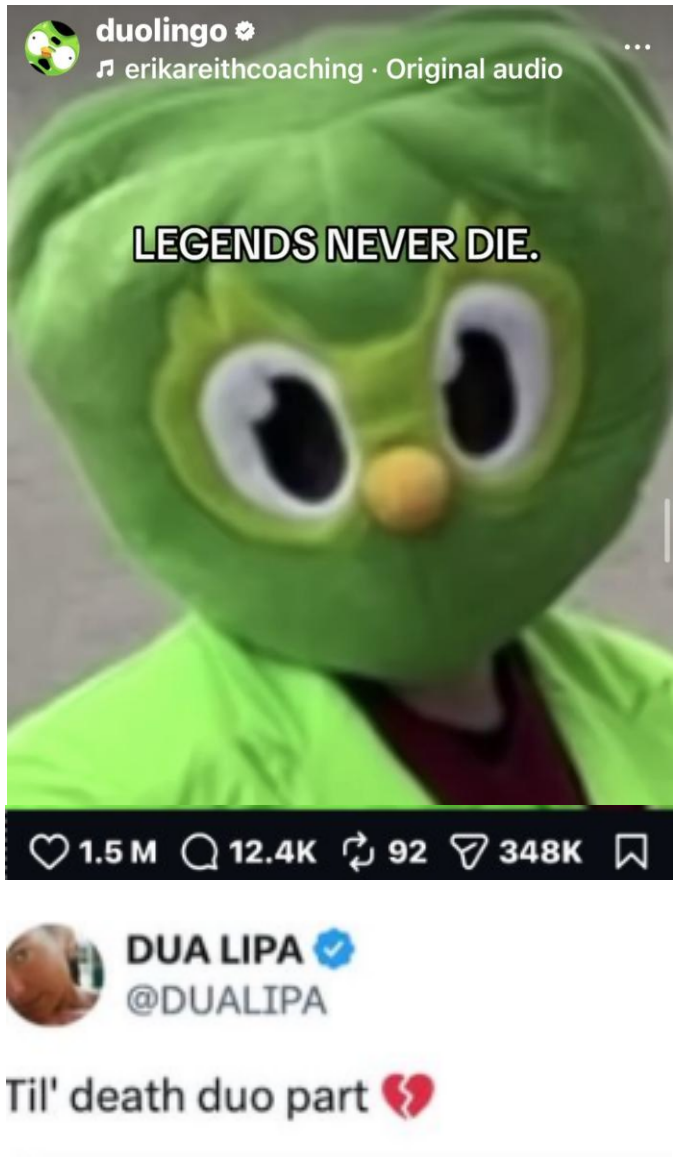
Juicy

Duolingo didn't always look like the app you see today. For years, it featured a light gray background, muted button colors, and a more robotic-looking Duo. But in 2018, a few of our designers and illustrators began working on a concept for a kids' app. With brighter colors, rounder corners, and a cuter, friendlier Duo, the new design made learning feel more like a game. It was exactly the kind of playful experience we wanted for the main app. So, even though we didn't expect it to boost metrics in the short-term, we took the leap and applied this new design language—dubbed Juicy—across all of Duolingo.

? ***DAUs:*** Daily Active Users. The number of individuals using Duolingo every day across iOS, Android, and the website.

"Death of Duo" Marketing Campaign





ENGAGEMENT IMPACT

7% rise in Subscriptions amongst users aged 18–24 in March 2025, suggesting the campaign's emotional impact supported stronger retention

5% spike in user growth and a **33%** increase in daily active usage during the campaign period

169,000 mentions of Duo within two weeks

25,560% spike in social media mentions on the day of the announcement

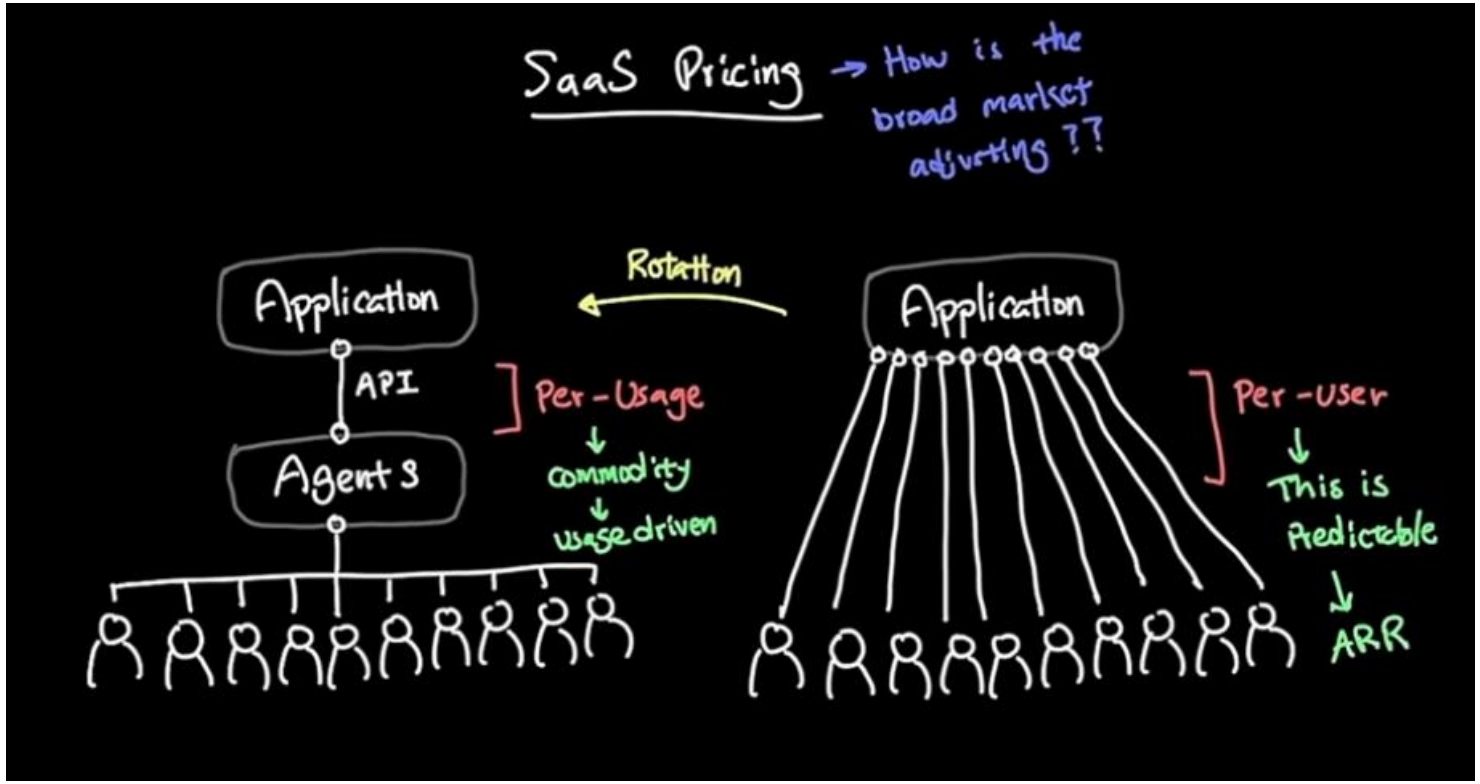
Failed Copy from Lipton







Commentary on SaaSpocalypse



https://www.youtube.com/watch?v=uY2AoCrrZ-s

The Timeline

- Jan 30: Anthropic launches plugins for Claude Cowork
- Feb 3: Market opens, traders point to Anthropic blog post
- Feb 5: Anthropic drops Claude Opus 4.6, head of enterprise product says "we are now transitioning almost into "vibe working"
- Feb 6: The term "vibe working" hits mainstream media, most publications pick up the news

Why it doesn't impact Duolingo

- The entire reason why vibe working is an issue is due to how SaaS pricing works – the market is worried that predictable ARR revenue isn't going to be predictable
- Instead, they think SaaS businesses are going to shift per-usage
- Duolingo doesn't have agents interact with the application – it's simply been brought down by the entire industry